



```
In [30]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```

```
In [31]: df = sns.load_dataset("titanic")
```

```
In [32]: df.head()
```

```
Out[32]:
```

|   | survived | pclass | sex    | age  | sibsp | parch | fare    | embarked | class | who   |
|---|----------|--------|--------|------|-------|-------|---------|----------|-------|-------|
| 0 | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | S        | Third | mar   |
| 1 | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | C        | First | womar |
| 2 | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | S        | Third | womar |
| 3 | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | S        | First | womar |
| 4 | 0        | 3      | male   | 35.0 | 0     | 0     | 8.0500  | S        | Third | mar   |

```
In [33]: df.columns
```

```
Out[33]: Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
               'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
               'alive', 'alone'],
              dtype='object')
```

```
In [34]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   survived        891 non-null    int64
1   pclass          891 non-null    int64
2   sex             891 non-null    object
3   age             714 non-null    float64
4   sibsp          891 non-null    int64
5   parch          891 non-null    int64
6   fare           891 non-null    float64
7   embarked        889 non-null    object
8   class           891 non-null    category
9   who             891 non-null    object
10  adult_male      891 non-null    bool
11  deck            203 non-null    category
12  embark_town     889 non-null    object
13  alive           891 non-null    object
14  alone           891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

```
In [35]: df.drop(["deck", "embark_town", "alive", "class", "who", "adult_male"], axis=1)
```

```
In [36]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   alone       891 non-null    bool
dtypes: bool(1), float64(2), int64(4), object(2)
memory usage: 56.7+ KB
```

```
In [37]: df["age"].fillna(df["age"].mean(), inplace=True)
```

```
In [38]: df.dropna(subset=["embarked"], inplace=True)
```

```
In [39]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 889 entries, 0 to 890
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    889 non-null    int64
1   pclass      889 non-null    int64
2   sex         889 non-null    object
3   age         889 non-null    float64
4   sibsp       889 non-null    int64
5   parch       889 non-null    int64
6   fare        889 non-null    float64
7   embarked    889 non-null    object
8   alone       889 non-null    bool
dtypes: bool(1), float64(2), int64(4), object(2)
memory usage: 63.4+ KB
```

```
In [40]: from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
```

```
In [41]: df.head()
```

```
Out[41]:
```

|   | survived | pclass | sex    | age  | sibsp | parch | fare    | embarked | alone |
|---|----------|--------|--------|------|-------|-------|---------|----------|-------|
| 0 | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | S        | False |
| 1 | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | C        | False |
| 2 | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | S        | True  |
| 3 | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | S        | False |
| 4 | 0        | 3      | male   | 35.0 | 0     | 0     | 8.0500  | S        | True  |

```
In [42]: df['sex'] = le.fit_transform(df['sex'])
df["embarked"] = le.fit_transform(df["embarked"]) # S=2, C=0, Q=1
```

```
In [43]: df = df.astype(int)
```

```
In [44]: df.head()
```

```
Out[44]:
```

|   | survived | pclass | sex | age | sibsp | parch | fare | embarked | alone |
|---|----------|--------|-----|-----|-------|-------|------|----------|-------|
| 0 | 0        | 3      | 1   | 22  | 1     | 0     | 7    | 2        | 0     |
| 1 | 1        | 1      | 0   | 38  | 1     | 0     | 71   | 0        | 0     |
| 2 | 1        | 3      | 0   | 26  | 0     | 0     | 7    | 2        | 1     |
| 3 | 1        | 1      | 0   | 35  | 1     | 0     | 53   | 2        | 0     |
| 4 | 0        | 3      | 1   | 35  | 0     | 0     | 8    | 2        | 1     |

```
In [45]: X = df.drop("survived", axis=1)
y = df["survived"]
```

```
In [46]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
In [47]: from sklearn.metrics import accuracy_score
```

```
In [48]: from sklearn.naive_bayes import GaussianNB
```

```
In [49]: model_NB = GaussianNB()
```

```
In [50]: model_NB.fit(X_train,y_train)
```

```
Out[50]:
```

▼ GaussianNB ⓘ ?

GaussianNB()

```
In [51]: y_pred_NB = model_NB.predict(X_test)
```

```
In [52]: y_pred_NB
```

```
Out[52]: array([0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1,
                0, 1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0,
                0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0,
                0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0,
                0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0,
                0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0,
                0, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0,
                0, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1, 0,
                0, 1])
```

```
In [53]: accuracy_score(y_test,y_pred_NB)
```

```
Out[53]: 0.7752808988764045
```

```
In [53]:
```